Alex Vermillion

<résumé@ajvermillion.com>

IL → AZ → MD → AZ

Interested in making complicated tasks simple and fallible tasks reliable.

Experience & Education

Computer Engineer @ U.S. Army "C5ISR" (Government Civilian)

06/2021–09/2026 (I left to move to AZ)

Testing and validation to bring a government cryptographic module through NSA certification as a KMI (Key Management Infrastructure) device.

Led (government side) of conversion of our certified solution from C to Rust (an NSA-approved Memory Safe Language) for an updated certification. Led meetings, set targets, built metrics, and removed blockers.

Read, annotated, and produced internal guides for NSA OTNK specifications. Attended conferences on related matters.

Used testing and development tools (NSA and consumer) and authored documentation and automation (mostly in Bash) to allow rapid and 'fool-proof' repeatability of common flows and actions.

Computer Science BSc @ Arizona State University

Honors Degree @ Barrett, the Honors College

09/2017–05/2021

Studied discrete mathematics, digital logic design, abstract theoretical computer science, and more. Supplemented classwork with private projects.

Capstone: "Psyche Driller" ↗

Created a simple game for the NASA Psyche mission. Used Agile methodology to design an easy-to-modify game with a group using the Godot language / engine. Gameplay involves piloting a drill around a procedurally-generated ore field, mining ores of different hardnesses and values to upgrade the drill (so you can do more mining).

Honors Thesis: "SSB and Whuffie" ↗

Studied reputation-metric design in Secure Scuttlebutt, a distributed social network.

"Rendering Bullets" @ ASU's Devil's Invent

11/2017

(18 hour hackathon)
Learned about the Human Interface Device operating mode for Raspberry Pi Zero prototyping boards, modified Open Source libraries to create the functionality needed, and wired up a programmable platform-agnostic wired controller capable of sending arbitrary keystrokes to any device with a USB A input. My team was awarded 1st place and $1,000 to continue our project.

"CamGuard" @ ASU's Devil's Invent

09/2017

(18 hour hackathon)
Used OpenWRT and a Raspberry Pi to create an IOT wireless access point with extensible security methods. My team was awarded 1st place and $1,000 to continue our project.

Skills

My limited public portfolio is available at AlexJVermillion.com. An overview of some of my potentially valuable skills is available below, but I really pride myself on my ability to learn.

Passionate About / Personal Projects

Rust	Creative Design		Programming
Haskell	Memory	Safe	HTML+CSS
Theoretical CompSci	Languages		C
Proofreading	Typesystem		

Familiar With / Can Use

C++	Python	Data Structures
Java	Algorithms	